Lab 3: Object Detection and Palletization

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***Abstract*— This lab focuses on implementing robotic object detection and palletization using the Dobot Magician Lite integrated with the YOLO framework. The objectives were to gain practical experience in computer vision–based object recognition, program motion sequences through Python and the pydobot library, and perform automated pick-and-place operations using a suction cup end effector. The Dobot utilized YOLO to detect and classify objects, then executed palletization by placing them into their respective baskets. The lab provided hands-on insight into vision-guided robotics, motion control, and the integration of machine learning with robotic manipulation.**

I. Problem Statement

Robotic palletization systems are widely used in industrial and research applications, but their effectiveness depends heavily on the robot's ability to accurately detect and classify objects. The challenge lies in integrating computer vision with robotic control to ensure that objects of varying shapes, orientations, and positions are correctly identified and placed in their designated locations.

This lab addresses these challenges by implementing YOLO-based object detection with the Dobot Magician Lite, which uses a suction cup end effector to pick and place objects into respective baskets. The goal is to evaluate the performance of vision-guided robotics in automating palletization tasks and to gain insight into the integration of machine learning with precise robotic motion control.

II. Procedure

All required packages and dependencies were installed on the Linux machine. We mounted the camera above the Dobot end effector as mentioned in the lab document. A Python script was used to calibrate the camera and test YOLO-based object detection, ensuring that objects were correctly identified with bounding boxes and labels. The suction cup end effector was attached to the Dobot, and a sample program was run to verify basic movements and suction control.

Once the setup was complete, objects were placed on the table within the camera's field of view, and baskets were fixed in position to drop the objects. A safe home position for the Dobot was defined in the Python script, along with an intermediate point for safe transitions between pick and place operations. YOLO was then used to detect the objects in real time, and their pixel coordinates were converted into the Dobot's workspace coordinates through calibration. Table 1 shows the coordinates of the baskets, home and pickup positions.

For each detected object, the Dobot was programmed to move from the home position to the object's location at a safe height, descend to the surface, and activate the suction pump to grip the object. The object was then moved to the corresponding basket based on the object's class and deactivated the suction pump to release the object. The robot returned to the intermediate point before continuing to the next detection.

The .suck() function controlled the suction cup during pick and drop operations, while the time.sleep() function was used to add delays to ensure stable gripping and releasing.

Table 1 (Positions)

	x (start)	y (start)	z (start)	r
Pickup Position	240	5	-42	0
Basket A	250	143	75	0
Basket B	-10	143	75	0
Home Position	240	-4	34	0

III. Results

The Dobot successfully executed the object detection and palletization tasks using YOLO. The Dobot demonstrated consistent accuracy when detecting objects, and placing them in their respective boxes. All objects were detected with >90% confidence rate.

The pixel density was increased by moving the end effector closer to the workspace, allowing YOLO to identify objects with high accuracy. The camera mount was locked to reduce vibration and eliminate tilt.

Overall, the Dobot was able to complete the intended operations, validating the programmed motion sequences while highlighting the importance of calibration and end effector selection for task-specific performance.

IV. Conclusion

This lab successfully demonstrated the integration of computer vision with robotic palletization using the Dobot Magician Lite. By combining YOLO-based object detection with suction cup pick-and-place operations, the system was able to recognize, classify, and sort objects into their respective baskets. The use of a calibrated camera setup enabled accurate mapping between detected pixel coordinates and the robot's workspace, while the programmed motion sequence ensured safe and repeatable operations.

The results highlight the potential of vision-guided robotics in automating industrial tasks such as sorting and palletization. Overall, the experiment provided valuable hands-on experience in integrating machine learning, computer vision, and robotic manipulation, and demonstrated how these technologies can be combined to create efficient and intelligent automation systems.

V. TECHNICAL SPECIFICATIONS

Robot Name: Dobot Magician Lite
Robot Serial Number: DT-MGL-4R002-01E
Repeatability: ±0.2mm
Maximum Payload: 0.25kg
Maximum Reach: 340mm
End Effector: Suction Cup ,Camera
Software Platform: Python
Safety Features: Collision Detection

VI. CODE AND VIDEO

Video for the object detection:
https://youtu.be/2WtvyThl0Vw

Repository Link:
https://github.com/VarunPro23/RAS/tree/main/Lab%203

Repository content:

- *Final_yolo.py* – code for Lab 3
- *Yolo_capture*.py – code to test yolo